COCREATE

Software Requirements Specification

Creator–Brand Marketing & Rewards Platform

Document Version: 1.0

MERN Stack Project

Initial Specification

1. Document Purpose

This Software Requirements Specification defines the functional, non-functional, technical, security, business and operational requirements for COCREATE.

COCREATE is a digital platform designed to connect brands/businesses with content creators through structured marketing campaigns. Brands can create campaigns, define content requirements, set creator eligibility criteria and establish rewards. Creators can discover campaigns, apply to participate, create and submit content, track campaign status and receive rewards for approved content. Platform administrators manage users, campaigns, submissions, disputes, transactions and overall platform operations.

2. Product Vision

COCREATE aims to provide a structured platform where brands and creators can collaborate, create meaningful content and build measurable marketing campaigns together.

The platform centralizes the workflow from campaign creation through creator selection, content submission, review, approval and reward management.

3. Problem Statement

Brand–creator collaborations often involve multiple disconnected processes for discovering creators, communicating campaign requirements, sharing assets, reviewing content, tracking performance and managing payments. This can lead to poor organization, unclear campaign status, manual communication and limited transparency.

COCREATE addresses this by providing a centralized creator–brand campaign marketplace and workflow management system.

4. Project Objectives

- Allow brands to create and manage marketing campaigns.
- Allow creators to discover relevant campaigns and apply.
- Provide structured creator selection and content review workflows.
- Provide transparent reward and transaction tracking.
- Provide secure authentication and role-based authorization.
- Give administrators control over users, campaigns, content and financial activity.
- Create a scalable MERN-based architecture suitable for future integrations.

5. Scope

5.1 In Scope

- User registration and authentication
- Creator profiles and social accounts
- Brand profiles
- Campaign creation and management
- Campaign discovery, search and filtering
- Creator applications

- Creator selection
- Content submission and review
- Revision requests
- Reward calculation
- Creator wallet and transaction history
- Notifications
- Admin dashboard and management
- Audit logging
- Responsive web interface

5.2 Out of Scope for MVP

- Advanced AI content analysis
- Automated fraud detection
- Full Instagram/TikTok API automation
- Complex tax management
- International settlement
- Native mobile applications
- Advanced AI creator recommendation
- Complex agency hierarchy
- Automated social-performance verification

6. User Roles

6.1 Creator

An individual who participates in campaigns and produces content.

- Register/login
- Manage profile
- Add social accounts
- Browse and filter campaigns
- Apply to campaigns
- Track applications
- Submit campaign content
- Respond to revision requests
- View rewards, wallet and transactions
- Manage notifications

6.2 Brand

A business or organization running creator marketing campaigns.

- Register/login
- Manage company profile
- Create/edit/publish campaigns
- Define creator and content requirements
- Review applications
- Accept/reject creators
- Review submissions

- Request revisions
- Approve/reject content
- Monitor campaign spending and performance

6.3 Administrator

A platform operator responsible for managing the COCREATE ecosystem.

- View platform dashboard
- Manage users, brands and creators
- Manage campaigns and submissions
- Monitor transactions and withdrawals
- Manage categories
- Suspend/reactivate accounts
- Review reported content
- View audit logs
- Manage platform settings

7. Authentication & Authorization Requirements

- FR-AUTH-001: Users shall register using name, email, password and account type.
- FR-AUTH-002: Admin accounts shall not be publicly registered.
- FR-AUTH-003: Email addresses shall be unique and validated.
- FR-AUTH-004: Passwords shall be securely hashed and never stored in plain text.
- FR-AUTH-005: Users shall be able to log in and log out securely.
- FR-AUTH-006: The system shall use short-lived access tokens and refresh tokens.
- FR-AUTH-007: Suspended or deactivated accounts shall not be permitted to perform restricted actions.
- FR-AUTH-008: Backend middleware shall enforce role-based authorization.

8. Creator Profile Requirements

- Name
- Profile photo
- Bio
- Location
- Creator category
- Languages
- Social accounts
- Follower count
- Portfolio
- Content types

Creator categories may include Fashion, Beauty, Food, Travel, Technology, Fitness, Gaming, Education, Lifestyle, Finance and Entertainment. Categories should be configurable by administrators.

9. Social Account Requirements

Creators may add Instagram, YouTube and TikTok accounts. For MVP, creators may manually provide platform, username, profile URL and follower count. Automated platform verification can be added in a future release.

10. Brand Profile Requirements

- Company name
- Logo
- Description
- Website
- Industry
- Location
- Contact information
- Social media links
- Campaign statistics

11. Campaign Management

Campaign management is the core feature of COCREATE. A campaign represents a marketing opportunity created by a brand.

11.1 Campaign Fields

- Campaign title
- Campaign description
- Category
- Social platform
- Campaign type
- Reward type
- Reward amount
- Total budget
- Start date
- End date
- Creator requirements
- Content requirements
- Campaign assets

11.2 Campaign Types

- Public Campaign: eligible creators can discover and apply.
- Private Campaign: participation is limited to invited creators; recommended for a later release.

11.3 Reward Types

- Per Post: fixed reward for each approved post.
- CPM: reward based on verified views per 1,000 impressions; recommended for V2.

11.4 Campaign Status

DRAFT → PENDING_REVIEW → PUBLISHED → ACTIVE → COMPLETED

Alternative statuses: PAUSED and CANCELLED.

12. Campaign Discovery

- Creators shall be able to browse active/public campaigns.
- Search shall support campaign title, brand name and category.
- Filters should include category, platform, reward, location, minimum followers and campaign status.
- Campaign detail pages shall show requirements, reward information, dates, eligibility and content guidelines.

13. Campaign Application

- FR-APP-001: Eligible creators shall be able to apply to a campaign.
- FR-APP-002: A creator shall not be able to submit duplicate applications to the same campaign.
- FR-APP-003: Applications shall contain campaign ID, creator ID, message, relevant social profile, date and status.
- FR-APP-004: Application statuses shall include PENDING, ACCEPTED, REJECTED and WITHDRAWN.
- FR-APP-005: Creators shall not apply to inactive or expired campaigns.
- FR-APP-006: Brands shall be able to view, accept and reject applications.

14. Content Requirements

Campaigns may specify content type, duration, format, hashtags, mentions, messaging, brand guidelines and submission deadlines.

- Video
- Image
- Reel
- Short-form video
- Other campaign-defined formats

15. Content Submission & Review

- FR-SUB-001: Accepted creators shall be able to submit campaign content.
- FR-SUB-002: Submissions shall support content URL, platform, caption, media and description.
- FR-SUB-003: Submission statuses shall include DRAFT, SUBMITTED, UNDER_REVIEW, APPROVED, REJECTED and REVISION_REQUESTED.
- FR-SUB-004: Brands shall be able to approve, reject or request revisions.
- FR-SUB-005: Rejection and revision requests should include a reason.
- FR-SUB-006: The system should preserve submission versions for auditability.
- FR-SUB-007: Brand/creator feedback related to a submission shall be stored.

16. Reward System

When a submission is approved, the system shall calculate the applicable reward, create a financial transaction and credit the creator's wallet.

For a per-post campaign, for example: Campaign reward = ₹1,500 → approved submission → creator wallet +₹1,500.

The system shall prevent duplicate reward crediting and shall ensure campaign spending does not exceed the configured budget.

17. Wallet & Transactions

17.1 Creator Wallet

- Available balance
- Pending balance
- Total earned
- Total withdrawn

17.2 Transaction Types

- REWARD
- WITHDRAWAL
- REFUND
- ADJUSTMENT
- PLATFORM_FEE

Financial transactions should be append-only from the application perspective and should not be physically deleted.

18. Withdrawal System

For MVP, withdrawals may be manually processed by administrators.

- Creator ID
- Amount
- Status
- Requested at
- Processed at
- Transaction reference

Withdrawal statuses: PENDING, PROCESSING, COMPLETED, REJECTED and FAILED.

19. Notifications

- Creator: application accepted/rejected, content rejected, revision requested, content approved, reward credited, withdrawal processed.
- Brand: new application, new submission, revision response, campaign completion.
- Admin: new brand registration, campaign requiring review, reported content, withdrawal request.

20. Admin Requirements

- Admin dashboard with platform KPIs

- User search/filter/view
- Creator and brand management
- Campaign moderation
- Submission moderation
- Transaction and withdrawal monitoring
- Category management
- Account suspension/reactivation
- Audit log access

21. Security Requirements

- Passwords shall be hashed using bcrypt or Argon2.
- Authentication credentials shall use secure transport (HTTPS in production).
- HTTP-only cookies should be used for sensitive token storage where applicable.
- Refresh-token rotation and session invalidation should be supported.
- All protected APIs shall require authentication.
- All role-sensitive operations shall be authorized server-side.
- Input validation and sanitization shall be applied.
- Rate limiting shall protect authentication and sensitive endpoints.
- CORS and security headers shall be configured appropriately.
- Secrets shall be stored in environment variables.
- Passwords and sensitive credentials shall never be written to logs.

22. Database Requirements

MongoDB with Mongoose will be the primary database.

Initial collections:

- users
- creatorProfiles
- brandProfiles
- campaigns
- campaignApplications
- submissions
- submissionComments
- wallets
- transactions
- notifications
- categories
- auditLogs

The database should use appropriate ObjectId references, indexes, timestamps, validation rules and soft deletion where required.

23. Suggested Core Data Models

23.1 User

Suggested fields: _id, name, email, passwordHash, role, isVerified, status, createdAt, updatedAt.

23.2 Campaign

Suggested fields: _id, brandId, title, description, category, platform, campaignType, rewardType, rewardAmount, totalBudget, remainingBudget, creatorRequirements, contentRequirements, status, startDate, endDate, createdAt, updatedAt.

23.3 Application

Suggested fields: _id, campaignId, creatorId, message, status, appliedAt, reviewedAt.

23.4 Submission

Suggested fields: _id, campaignId, creatorId, applicationId, contentUrl, platform, caption, media, status, rewardAmount, version, submittedAt, reviewedAt.

24. API Requirements

The backend shall expose REST APIs grouped by domain.

- /api/auth
- /api/users
- /api/creators
- /api/brands
- /api/campaigns
- /api/applications
- /api/submissions
- /api/wallet
- /api/transactions
- /api/notifications
- /api/admin

25. Initial API Catalogue

Method	Endpoint	Purpose
POST	/api/auth/register	Register creator or brand
POST	/api/auth/login	Authenticate user
POST	/api/auth/logout	Terminate session
POST	/api/auth/refresh	Refresh access token
GET	/api/campaigns	Browse campaigns
POST	/api/campaigns	Create campaign
GET	/api/campaigns/:id	Get campaign details
PATCH	/api/campaigns/:id	Update campaign
DELETE	/api/campaigns/:id	Delete/archive campaign
POST	/api/campaigns/:id/applications	Apply to campaign
GET	/api/applications/my	Creator applications
POST	/api/submissions	Submit content
GET	/api/submissions/my	Creator submissions

PATCH	/api/submissions/:id	Update/review submission
GET	/api/wallet	Get creator wallet
GET	/api/transactions	Get transaction history

26. Frontend Requirements

26.1 Public Pages

- Landing page
- Campaign discovery
- Campaign details
- About
- Login
- Registration
- Forgot password

26.2 Creator Pages

- Dashboard
- Profile
- Campaigns
- Campaign details
- Applications
- Active campaigns
- Submissions
- Wallet
- Transactions
- Notifications
- Settings

26.3 Brand Pages

- Dashboard
- Company profile
- Campaigns
- Create campaign
- Campaign details
- Applications
- Creators
- Submissions
- Analytics
- Settings

26.4 Admin Pages

- Dashboard
- Users
- Creators
- Brands
- Campaigns

- Submissions
- Transactions
- Withdrawals
- Categories
- Audit logs
- Settings

27. UI/UX Requirements

- Modern SaaS-style interface
- Clean layouts and consistent spacing
- Clear calls to action
- Status badges
- Campaign cards
- Data tables for administrative views
- Charts for analytics
- Responsive layouts
- Mobile-friendly creator workflows
- Consistent design system and reusable components

28. Responsive Requirements

The platform shall support desktop, laptop, tablet and mobile screen sizes. Critical creator actions such as campaign browsing, applications, content submission, earnings and notifications should be usable on mobile.

29. Performance Requirements

- Use pagination for large datasets.
- Use appropriate MongoDB indexes.
- Avoid unnecessary API requests.
- Optimize uploaded media.
- Lazy-load frontend modules where appropriate.
- Use caching where it materially improves performance.
- Target normal API responses under approximately 500 ms, excluding third-party service delays, under normal operating conditions.

30. File Management

Cloudinary is recommended for profile images, brand logos, campaign assets and creator media. The database should store file references/metadata rather than large binary files.

- Cloudinary URL
- Public ID
- File type
- File size
- Uploaded by
- Uploaded at

31. Audit Logging

Important actions should be recorded for accountability and troubleshooting.

- User
- Action
- Entity
- Entity ID
- Timestamp
- Relevant metadata

Examples include campaign creation, creator acceptance, content approval and account suspension.

32. Non-Functional Requirements

32.1 Maintainability

- Modular architecture
- Reusable components
- Service layer separation
- Clear naming conventions
- Centralized configuration
- Consistent error handling

32.2 Scalability

The architecture should be capable of scaling from an initial small user base to thousands of users without requiring a complete rewrite.

32.3 Availability

Production deployment should target reliable service availability with monitoring and appropriate recovery procedures.

33. Recommended Technology Stack

- Frontend: React + Vite + React Router + Tailwind CSS + Axios + Zustand + React Hook Form + Zod + Recharts
- Backend: Node.js + Express.js + MongoDB + Mongoose + JWT + bcrypt/Argon2 + Redis
- File storage: Cloudinary
- Email: transactional email provider
- Payments: payment gateway in V2
- Social APIs: official platform APIs in V2

34. Deployment Architecture

Recommended architecture:

Internet → React Frontend → HTTPS → Node/Express REST API → MongoDB / Redis / Cloudinary

The frontend may be deployed on a static hosting service and the backend on a cloud VM such as AWS EC2. Production must use HTTPS and environment-specific secrets.

35. Development Phases

Phase	Scope
Phase 1 — Foundation	Project setup, Git repository, frontend/backend setup, MongoDB, environment configuration and base architecture.
Phase 2 — Authentication	Registration, login, logout, JWT, refresh tokens, password hashing, email verification and RBAC.
Phase 3 — Creator	Creator profile, social accounts and creator dashboard.
Phase 4 — Brand	Brand profile and brand dashboard.
Phase 5 — Campaign	Create/edit/publish campaigns, discovery, search, filters and campaign details.
Phase 6 — Applications	Creator applications and brand acceptance/rejection.
Phase 7 — Submissions	Content submission, review, approval, rejection and revision workflow.
Phase 8 — Rewards	Reward calculation, wallet and transactions.
Phase 9 — Admin	Admin dashboard and management tools.
Phase 10 — Testing	Unit, API, integration, frontend, security and authorization testing.
Phase 11 — Deployment	Production frontend/backend, database, domain, HTTPS, environment variables and monitoring.

36. MVP Definition

The COCREATE MVP is complete when a brand can register, create and publish a campaign; a creator can register, complete a profile, discover the campaign and apply; the brand can accept the creator; the creator can submit content; the brand can approve it; and the system can generate a reward and reflect it in the creator wallet.

37. Future Roadmap

- V1.1: advanced filtering, richer analytics, email notifications, campaign invitations and submission version history.
- V2: payment gateway, automated withdrawals, official social integrations, CPM rewards and campaign analytics.
- V3: AI creator recommendations, fraud detection, automated content/performance verification, agency accounts and advanced reporting.

38. Key Business Rules

- BR-001: A creator cannot apply to an inactive or expired campaign.
- BR-002: A creator cannot submit content unless their application has been accepted.
- BR-003: A creator cannot submit duplicate content where campaign rules prohibit it.
- BR-004: A brand can manage only campaigns belonging to its organization/account.
- BR-005: Only authorized brand users can approve submissions.

- BR-006: Rewards are generated only after qualifying content is approved.
- BR-007: The same submission cannot generate the same reward twice.
- BR-008: Campaign spending cannot exceed the configured budget.
- BR-009: Suspended users cannot participate in campaigns.
- BR-010: Financial transaction records must not be physically deleted.
- BR-011: Administrative permissions must be verified server-side.
- BR-012: Approved submissions cannot be modified without an authorized revision workflow.

39. MVP Acceptance Criteria

- A creator can register, authenticate and manage a profile.
- A brand can register, authenticate and manage a company profile.
- A brand can create and publish a campaign.
- A creator can browse, search and view campaigns.
- A creator can apply to an eligible campaign.
- A brand can accept or reject applications.
- An accepted creator can submit campaign content.
- A brand can review, approve, reject or request revisions.
- An approved submission can generate a reward.
- The reward is reflected in the creator wallet and transaction history.
- Admin can monitor and manage core platform entities.
- Unauthorized users cannot access protected resources.
- The application is responsive and deployable.

40. Developer Implementation Principle

COCREATE should be developed as a workflow-driven application rather than as a collection of unrelated CRUD screens. Each major requirement should map to a database model, API endpoint, business rule, frontend component/page, validation rule and acceptance test.

Recommended implementation order: Requirements → User flows → Database design → Authentication → Campaigns → Applications → Submissions → Rewards → Wallet → Admin → Frontend refinement → Integrations → Testing → Deployment.

41. Suggested Next Specification Documents

- Part 2: Detailed Functional Requirements Catalogue (FR-AUTH, FR-USER, FR-CREATOR, FR-BRAND, FR-CAM, FR-APP, FR-SUB, FR-REWARD, FR-WALLET, FR-ADMIN).
- Part 3: Complete MongoDB schema and relationship specification.
- Part 4: Complete REST API specification with request/response schemas, validation and authorization.
- Part 5: Frontend information architecture, page-by-page requirements and component map.
- Part 6: Development task breakdown and Git milestone plan.